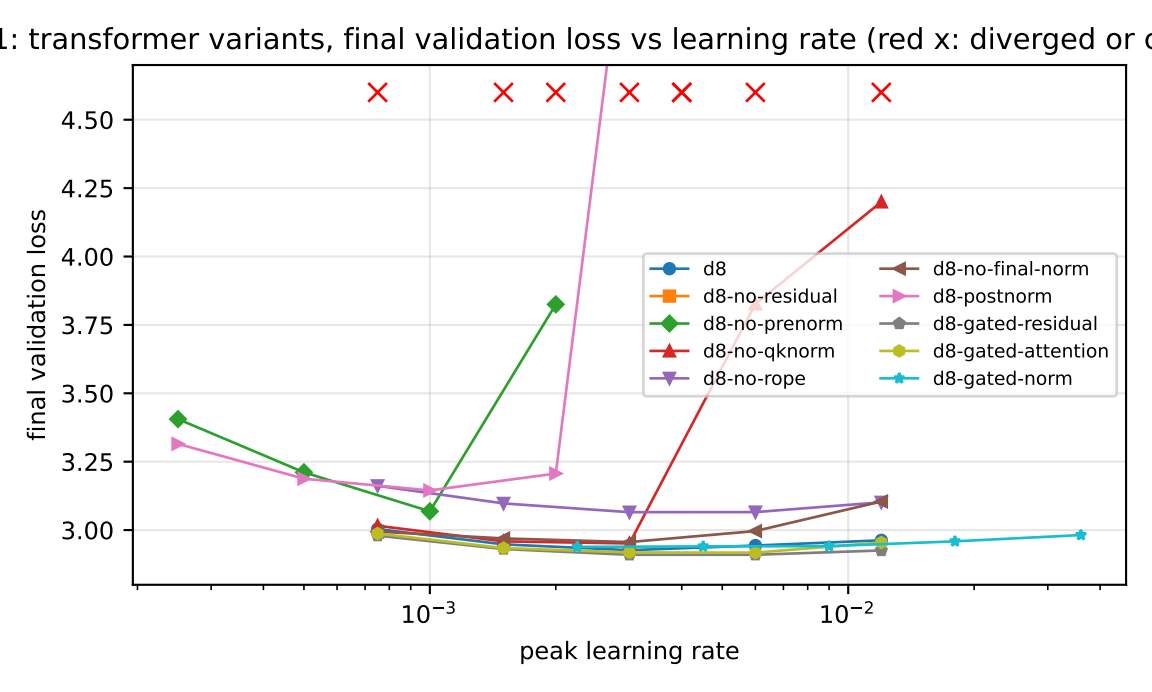
1: transformer variants, final validation loss vs learning rate (red x: diverged or c
final validation loss
4.50
4.25
4.00
3.75
3.50
3.25
3.00
10^{-3}
10^{-2}
peak learning rate
d8
d8-no-residual
d8-no-prenorm
d8-no-qknorm
d8-no-rope
d8-no-final-norm
d8-postnorm
d8-gated-residual
d8-gated-attention
d8-gated-norm